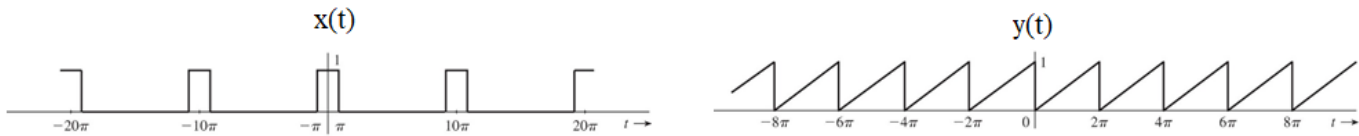


ENGR 3323: Signals and Systems

HW 9_Ch6

Q1) For the periodic signals $x(t)$ and $y(t)$ shown below:



- Find the exponential Fourier series for $x(t)$ and $y(t)$.
- Sketch the amplitude and phase spectra for signal $x(t)$.
- Use Parseval's theorem to approximate the power of the periodic signal $x(t)$ by calculating the power of the first N^{th} harmonics, such that the strength of the N^{th} harmonic is 10% or more of the power of the DC component.

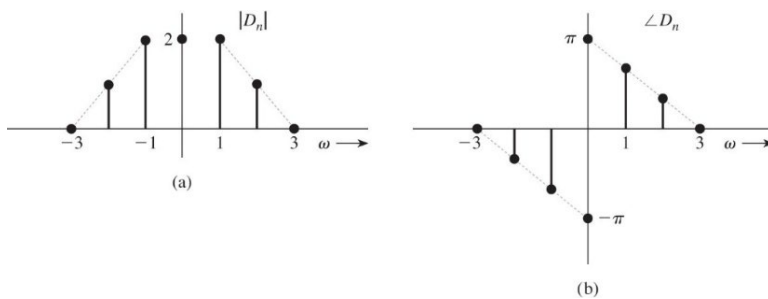
Q2) The exponential Fourier series of a certain function is given as

$$x(t) = (2 + j2)e^{j3t} + j2e^{jt} + 3 - j2e^{jt} + (2 - j2)e^{j3t}$$

- Sketch the exponential Fourier spectra.
- By inspection of the spectra in part (a), sketch the trigonometric Fourier spectra for $x(t)$. Find the compact trigonometric Fourier series from these spectra.
- Find the trigonometric Fourier series for $x(t)$.
- Find the signal bandwidth.

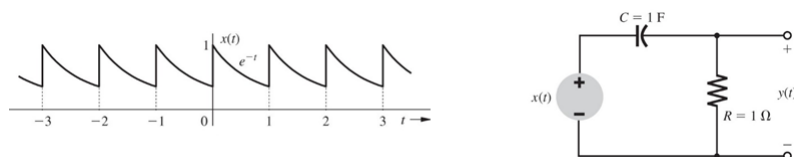
Q3) Figure below shows the exponential Fourier spectra of a periodic signal $x(t)$.

- By inspection of the Figure find the exponential Fourier series representing $x(t)$.
- By inspection of the Figure, sketch the trigonometric Fourier spectra for $x(t)$.
- By inspection of the trigonometric Fourier spectra found in part (b), find the trigonometric Fourier series for $x(t)$.



Q4) Find the response of an LTIC system with transfer function $H(s) = \frac{s}{s^2 + 2s + 3}$ to the periodic input $x(t) = (2 + j2)e^{j3t} + j2e^{jt} + 3 - j2e^{jt} + (2 - j2)e^{j3t}$. Express output in compact trigonometric FS.

Q5) Find the exponential Fourier series for a periodic signal $x(t)$ shown in Figure below



The signal $x(t)$ is applied at the input of an LTIC system shown above. Find the expression for the output $y(t)$.